COMP 3000 – Official project initiation document

By Will Keogh

Project title: Digital wellbeing companion

Introduction:

The project I am presenting is a digital wellbeing companion that acts as an online deterrent for people who cannot control their computer screen habits. This platform will be designed and created for people who need support digitally, users who might game for extended periods of time, users who might not know when to take a break while doing online work and users who simply want an easy way to motivate them to keep up good screen time habits.

What is the problem being solved?

The problem that I am combating in this project mostly stems from the issue of screen addiction, and living in the age that we do there will be people out there who need help monitoring how much of their time digitally. Anyone who would like to help cut down on how much time they spend looking at a screen will be the target user for this project. Gaming addiction is a good example, so people playing throughout the day and night with no breaks can lead to major health risks. They won't take care of themselves, physically and mentally, as well as becoming massively unproductive. But just like any addiction, this must be tackled carefully and in a positive way.

Project details and features:

As stated before, my project is called 'Digital wellbeing companion' and is an application that would run in the background initially of a user's computer like an onscreen overlay. It would then act to help encourage users to manage their screen time well and stay healthy. Such as using onscreen pop-up messages to remind users how long they have been going for, make suggestions of other activities they could do or even just suggest taking a 5-minute tea break. Users could set goals on the app, such as how much time they will allow themselves to play this game, etc. These will not include parental controls, and the aim of the product is too be managed by the user as I would like all features to be completely voluntary. And screen time limiters never work.

The app would not punish the user for missing their goals but instead reward them for following it, with encouragement to hit it more often, The app would offer tools such as a point system that users could then use to buy in app customization items and allow for a friendly scoreboard with other users, friendly messages to help keep the user feeling positive about what they are doing and links to online resources that offer help to people struggling with online addiction or similar problems. A streak system could also be implemented; this also being used to help encourage users to stick to their screen time goals. Similar to apps like Duolingo, it is a simple way to encourage users to keep up app usage. If I have enough development time, the app will feature simple puzzle games that users can take part in to help create breaks between intensive gaming or work sessions. This will be an easy stress release strategy.

This project aims to be a positive force that will help people stay healthy online. Helping to minimize the health issues that come with too much time on a computer as well as mental strain. Hopefully helping to reduce the stress users feel from online gaming as well as other digital activities. It will not force people to stop gaming or act like a strict parent but will become a virtual buddy or coach to help and support users with poor screentime management.

Technical details:

The project development will use different technical techniques to create each part of the app. The application will feature a secure login system which will be required if users are to save their data across different hardware and allow for multiple people to use the app. I will use databases to store all the relevant data in complacence with GDPR laws. The data will have to be encrypted to keep it secure, and users will be provided with a term of service, so they are aware of what of their data will be stored.

I am planning on creating the project in Microsoft Visual studio using Typescript as it is more secure than regular Java. This will be the main language used for most of the project's development. The databases used for data storage will be created using SQL Lite as it is the one I am most familiar with due to previous experience. I am yet to decide on an encryption method, but it will be one that is compatible with the databases and the type of user data I am storing, so usernames, passwords and emails.

The app will use surface level monitoring to measure how long a user has had a game open for. This is to ensure it doesn't be seen as a virus or unwanted third-party software. There is also the risk that it could violate a privacy policy if it was to be integrated into game launches like Steam or Epic games. Examples when comparing it to other applications include Nvidia shadow play or Overwolf, I intend it to function similarly.

Conclusion:

In conclusion, this document does not represent the final product, and I will detail any changes made during development in my report. I hope for it to be a successful project and for the final product to contain all the features I originally proposed.